Tsz Fei Wang

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Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages: Python, Java, C, C++, TypeScript, JavaScript, HTML/CSS, SQL, R, GLSL
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Three.js, PyTorch, NumPy, Pandas, Qt/QML
Tools & Platforms: Git, Docker, GitHub Actions, Firebase, Elasticsearch, Vercel, Cloudflare Workers, Postman, Jupyter, KiCad

Experience

Century Frontier Asset Management — Jul 2026 – Aug 2026
Quantitative Research Intern — *Shenzhen, CN*

- Built a point-in-time backtesting pipeline in **Python** and **pandas** that eliminated lookahead bias from the firm's LLM-driven equity research across **104 tickers**.
- Benchmarked **9 LLM architectures** on stock-preference and inter-stock-relationship prediction, showing that a model's stated confidence was a poor predictor of its accuracy.

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*

- Built most of the frontend for Catrin's full-stack data access portal, including the search, filter and results views researchers use to explore **~4 million** TF-TG associations across **33,000 genes**.
- Reconciled the schemas of **9 independent data sources** so the portal can query them as one dataset.

Mackenzie Cybersecurity Team — Jun 2025 – Jun 2026
Team Captain, President — *Toronto, ON*

- Captained the team to **First in Canada** at CyberTitan, Canada's largest cyberdefense competition, hardening Windows, Linux and Cisco systems under time pressure.
- Created and taught a cybersecurity curriculum reaching **60+ students** across the school.

Non-Trivial Research Fellowship — Mar 2025 – Aug 2025
Machine Learning Research Fellow — *Remote*

- Trained and evaluated a **PyTorch** audio deepfake classifier in **Python** on a Kaggle dataset of real and AI-synthesized speech.
- Researched AI voice cloning as an emerging security threat and proposed a detection-based defence in a written research proposal.

Projects

Technova Hackathon Badge | *KiCad, Hardware Lead* — *Oct 2026*

- Led hardware design of a custom NFC badge for a **~300-person** hackathon in **KiCad**, taking it from schematic to a manufacturing-ready JLCPCB assembly order.
- Cut the per-badge cost from about **$11 to $7** by reworking the parts list around in-stock LCSC components, while fitting NFC check-in, an OLED display, LEDs and USB-C charging.

Submerge | *GLSL, QML, Python* — *Oct 2026 – Present*

- Shipped an animated KDE Plasma theme in **GLSL** and **QML** downloaded **~1,290 times** on the KDE Store, with a shader water simulation and a live system HUD.
- Resolved a user-reported NVIDIA/Wayland rendering failure by porting the shaders to GLSL ES, and added power-aware frame pacing to save battery.

Physical Studio | *Python, React, Three.js, Docker* | ***Winner**, Hack the Valley 11* — *Oct 2026*

- Won Hack the Valley 11 with a tool that turns a prompt into a step-by-step hardware build guide; engineered the **Python** MCP server that lets an LLM design from measured parts and check its own work for collisions.
- Built the **React Three Fiber** viewer that animates each assembly step and deployed the app with **Docker Compose** on Vultr.

Awards & Honors

Gold & Silver Medalist, IT Systems Networking, Skills Ontario (Regional & Provincial) — Mar 2026 – May 2026
National Champion x2, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
National Finalist, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025